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Backup/Restore Tools performance Comparison

Vinicius Grippa

Lead Senior Support Engineer at Percona

Vinicius Grippa is a Percona Senior Support Engineer, Oracle Ace, and author of the book Learning MySQL. Vinicius has a Bachelor's degree in Computer Science and has been working with databases for 16 years. He has experience in designing databases for mission-critical applications and, in the last few years, has become a specialist in MySQL and MongoDB ecosystems. Working in the Support team, he has helped Percona customers with hundreds of different cases featuring a vast range of scenarios and complexities. Vinicius is also active in the OS community, participating in virtual rooms like Slack, and speaking at MeetUps, and presenting conferences in Europe, Asia, North and South America.

Backup Tools

Backup Tools

- mysqldump
- mydumper
- mysqlpump
- Xtrabackup
- MySQL shell

Backup Commands

```
mysqldump  --all-databases  --max-allowed-packet=4294967295 --single-
transaction -R --master-data=2 --flush-logs | gzip  > /backup/dump.dmp.gz

mydumper  --threads=16  --trx-consistency-only --events --routines --triggers -
-compress --outputdir /backup/ --logfile /backup/log.out --verbose=2
```

Backup Commands

```
util.dumpInstance("/backup/mysqlshell", {ocimds: true, compatibility:
["strip_restricted_grants","ignore_missing_pks"],threads: 16})
```

```
mysqlpump --compression-algorithms=zstd --default-parallelism=16 --all-
databases > backup.out
```

```
xtrabackup --backup --parallel=32 --compress --compress-threads=16 --
datadir=/mysql/ --target-dir=/backup/xtrabackup/compress/
```

Restore Commands

```
# same for mysqldump and mysqlpump
$ time mysql < dump.out


# myloader no compression 16 threads
$ time myloader --threads=16 --host=127.0.0.1 --port=3306 --
directory=/backup/mydumper/no_compress --overwrite-tables
```

Restore Commands

```
$ time xtrabackup  --decompress --parallel=32 --target-dir=/backup/xtrabackup/compress
$ time xtrabackup --prepare  --target-dir=/backup/xtrabackup/no_compress/
$ time cp no_compress/* /mysql/ -R

mysql> util.loadDump("/backup/mysqlshell", {resetProgress:true, progressFile
:"/backup/restore.json",threads: 6})
```

Backup Options

- Physical Backups - Copies all the physical files that belongs to the database.
- Logical Backup - You don't take copies of files, you only extract data from the data files into dump files.

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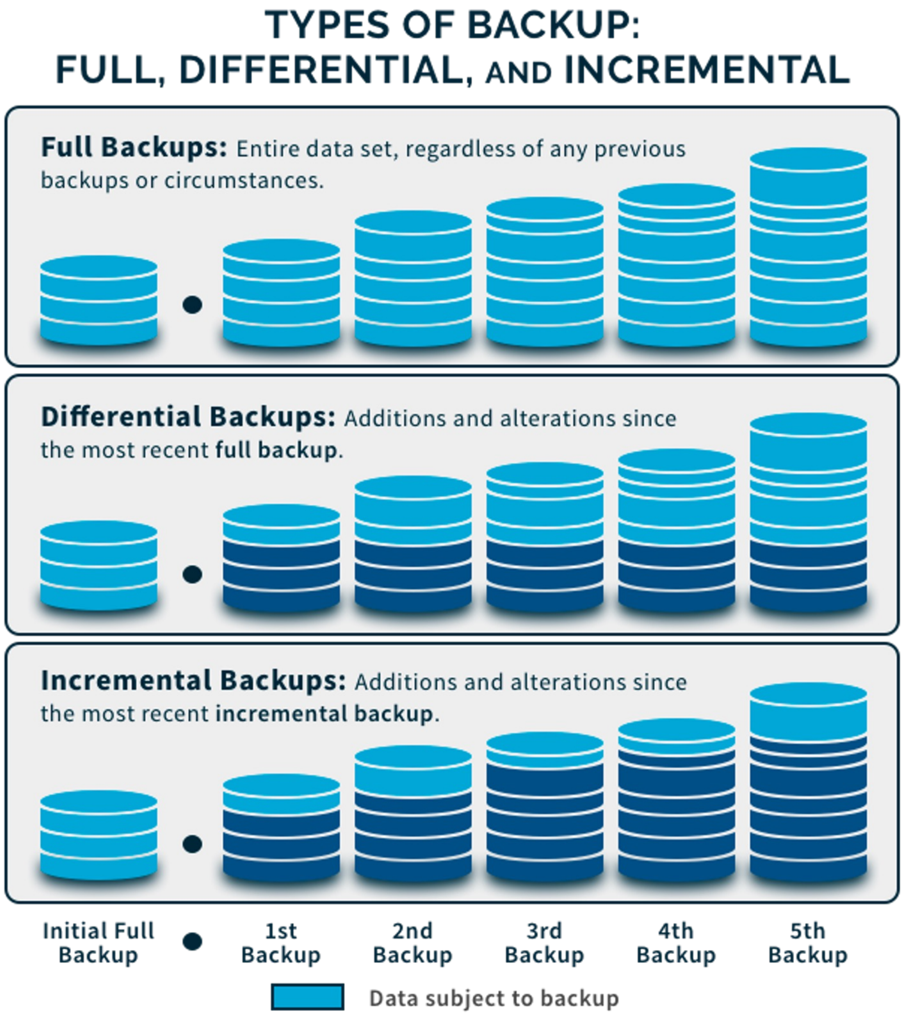
Backup Options



Things to consider when taking a backup

- Network Encryption
- File encryption
- Single Thread x Multi Thread
- Data compression
- Backup Size
- Backup Time
- Restore Time
- Total time
- PITR
- etc...

Backup Benchmark

The first side of the history…

Benchmark

Specs of the benchmark:

- 32 CPUs (m5dn.8xlarge)
- 128GB Memory
- 2x io1 disks 600GB and 5000 IOPs
- Centos 7.9

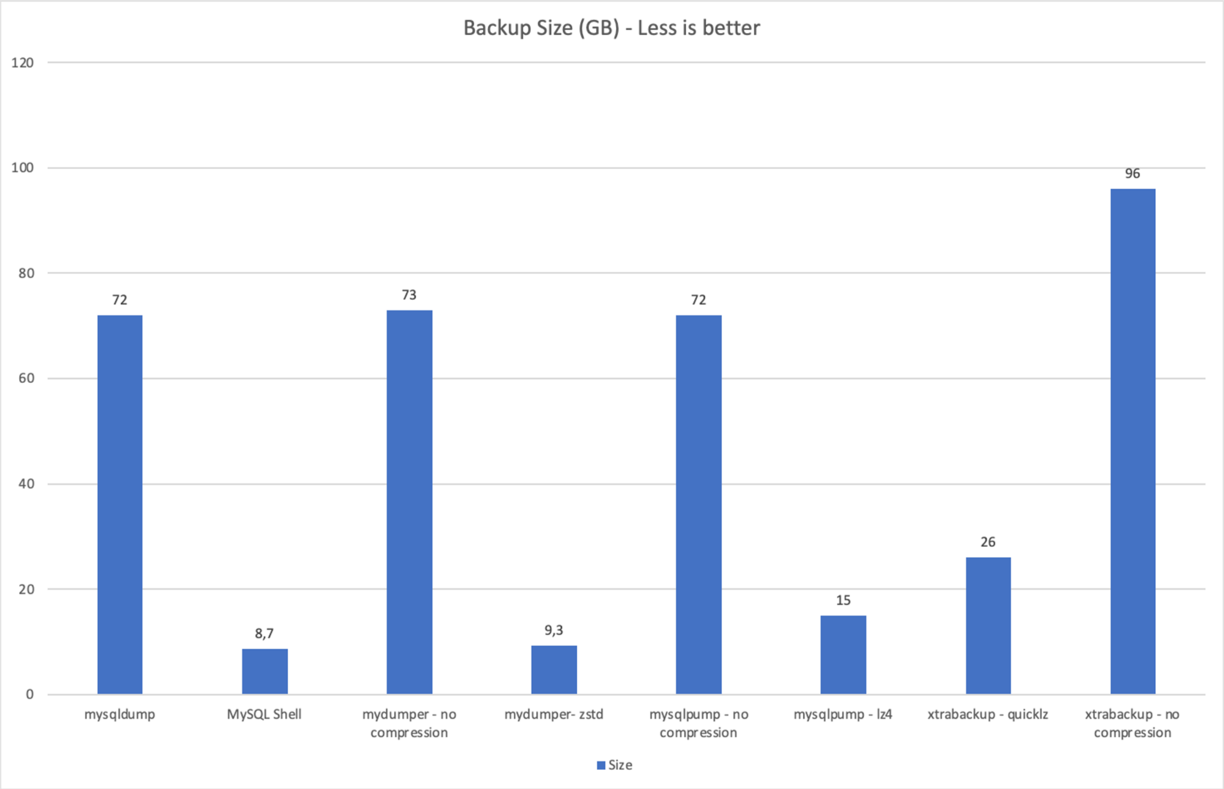
Database:

- 96GB (In-Disk)
- 90 Tables
- Different sizes
- tpcc (sysbench)

Tool Version:

- MySQL 8.0.26
- MySQL shell 8.0.26
- mydumper 0.11.5 – gzip
- mydumper 0.11.5 – zstd
- Xtrabackup 8.0.26
- tpcc (sysbench)

Results



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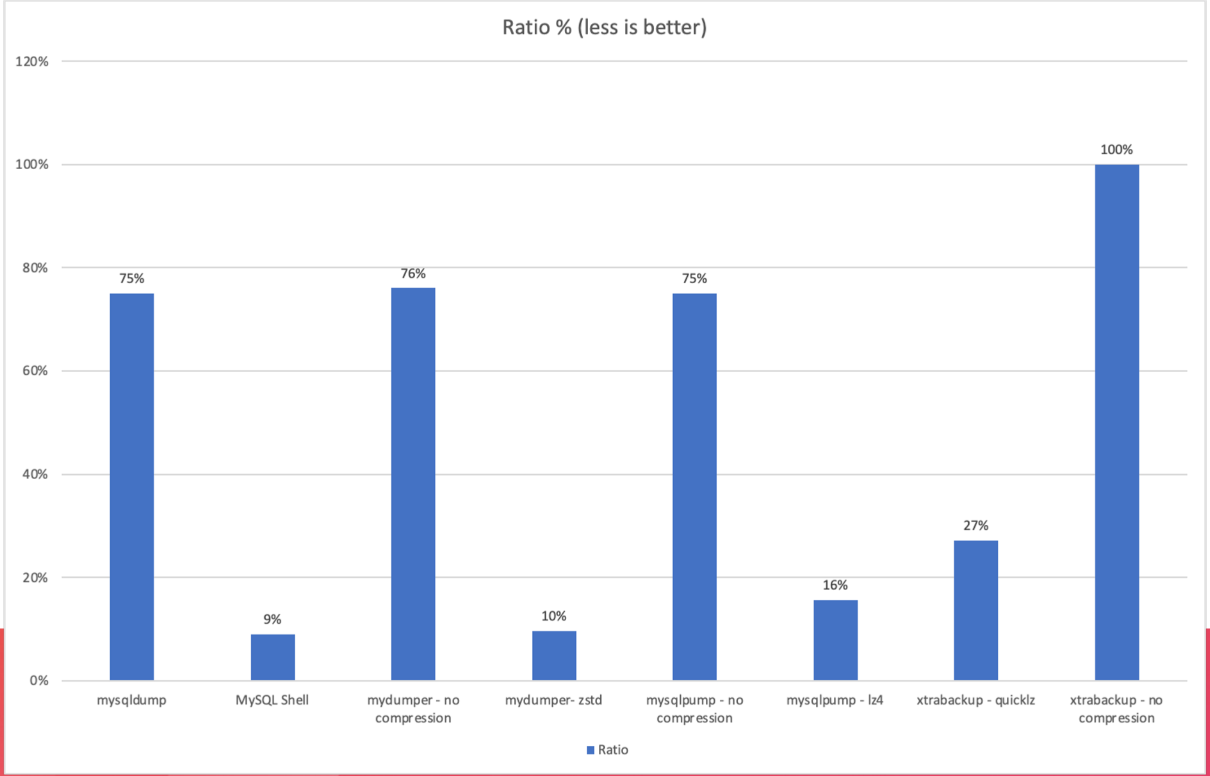
Results



Ratio % (less is better)
120%
100%
80%
60%
40%
20%
0%
75%
9%
76%
10%
75%
16%
27%
100%
mysqldump
MySQL Shell
mydumper - no compression
mydumper- zstd
mysqlpump - no compression
mysqlpump - lz4
xtrabackup - quicklz
xtrabackup - no compression
Ratio

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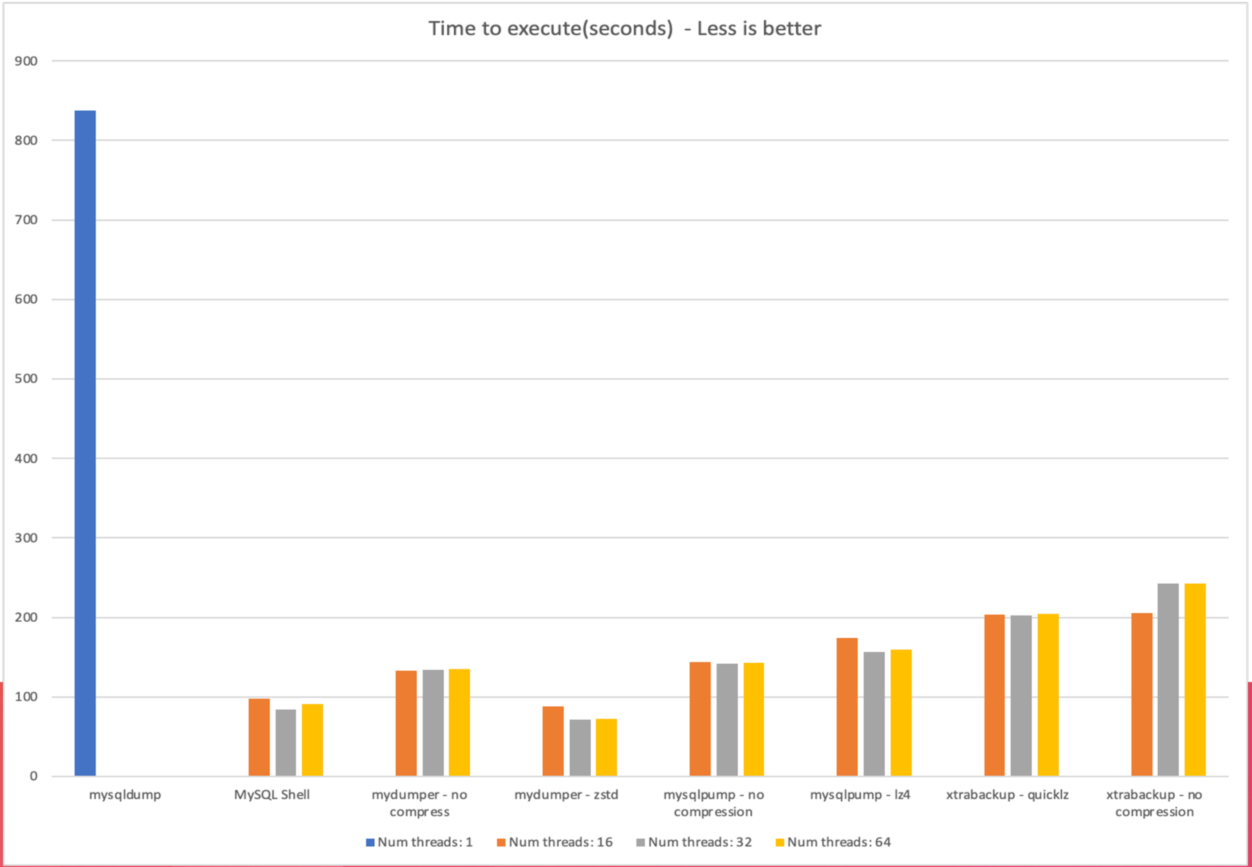
Results



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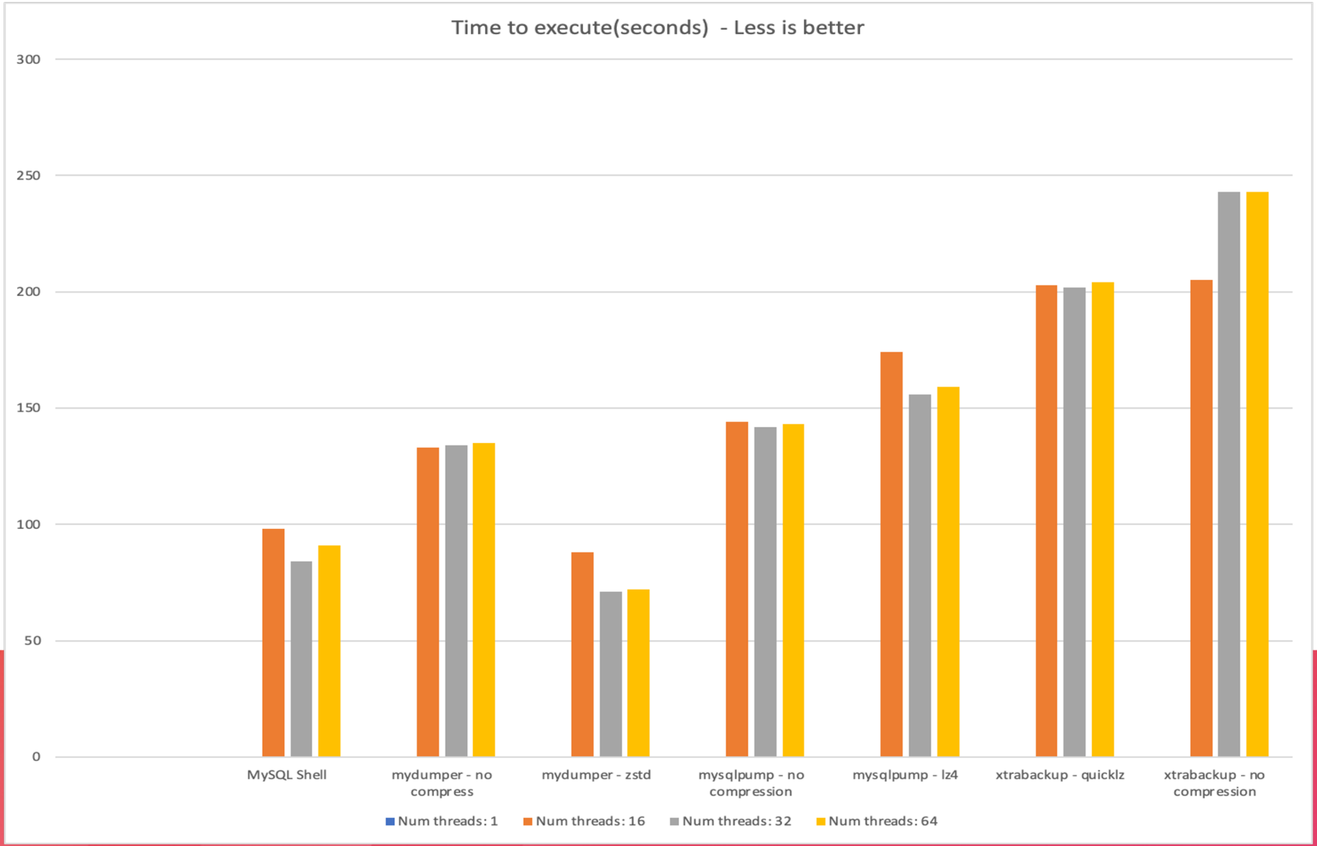
Results


Time to execute(seconds) - Less is better
300
250
200
150
100
50
0
MySQL Shell
mydumper - no compress
mydumper - zstd
mysqlpump - no compression
mysqlpump - lz4
xtrabackup - quicklz
xtrabackup - no compression
Num threads: 1
Num threads: 16
Num threads: 32
Num threads: 64

Questions and comments

- Does xtrabackup support zstd?
  - A: No, but we opened a FR:
  - https://jira.percona.com/browse/PXB-2669
  - You can "pipe" xtrabackup output
- TCP/IP vs Socket?
  - A: The impact is neglectable.
- SSL vs No SSL?
  - Compared both options with mydumper and it does not seem a big difference:
  - SSL: 76s
  - No SSL: 72s

Restore Benchmark

The second side of the history...

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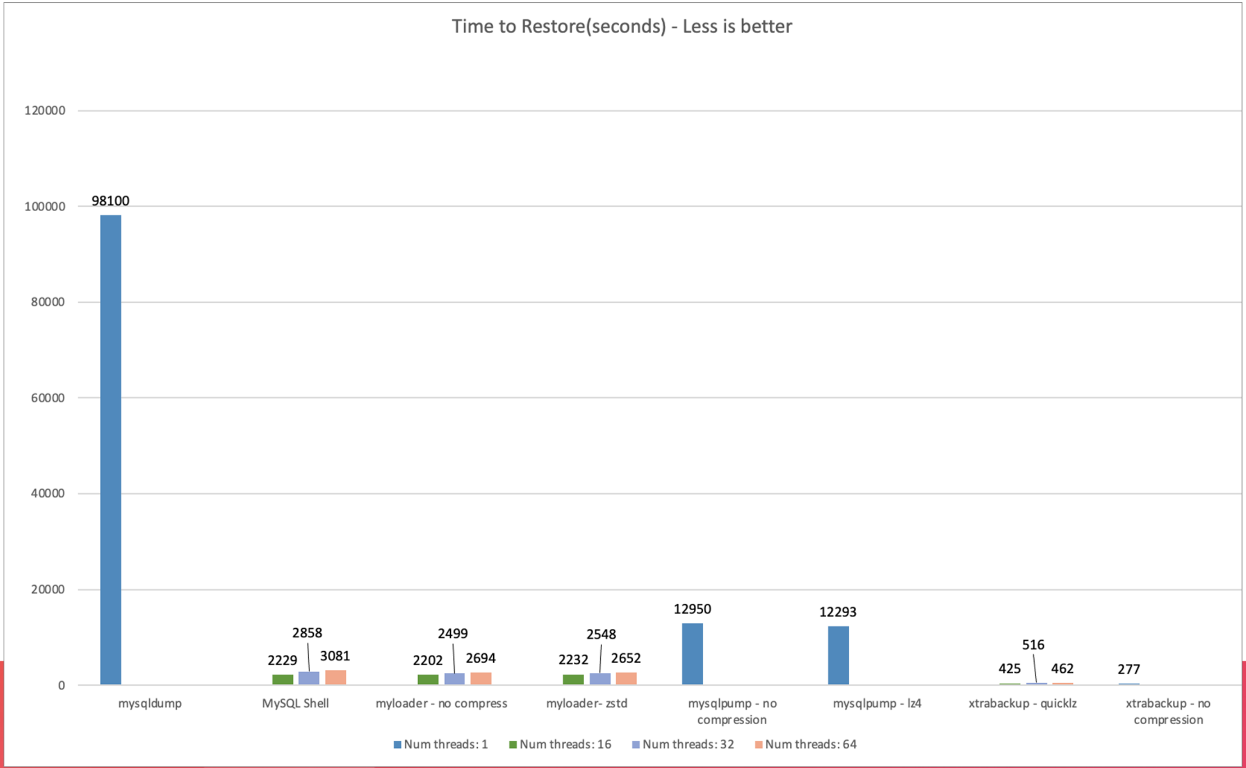
Results



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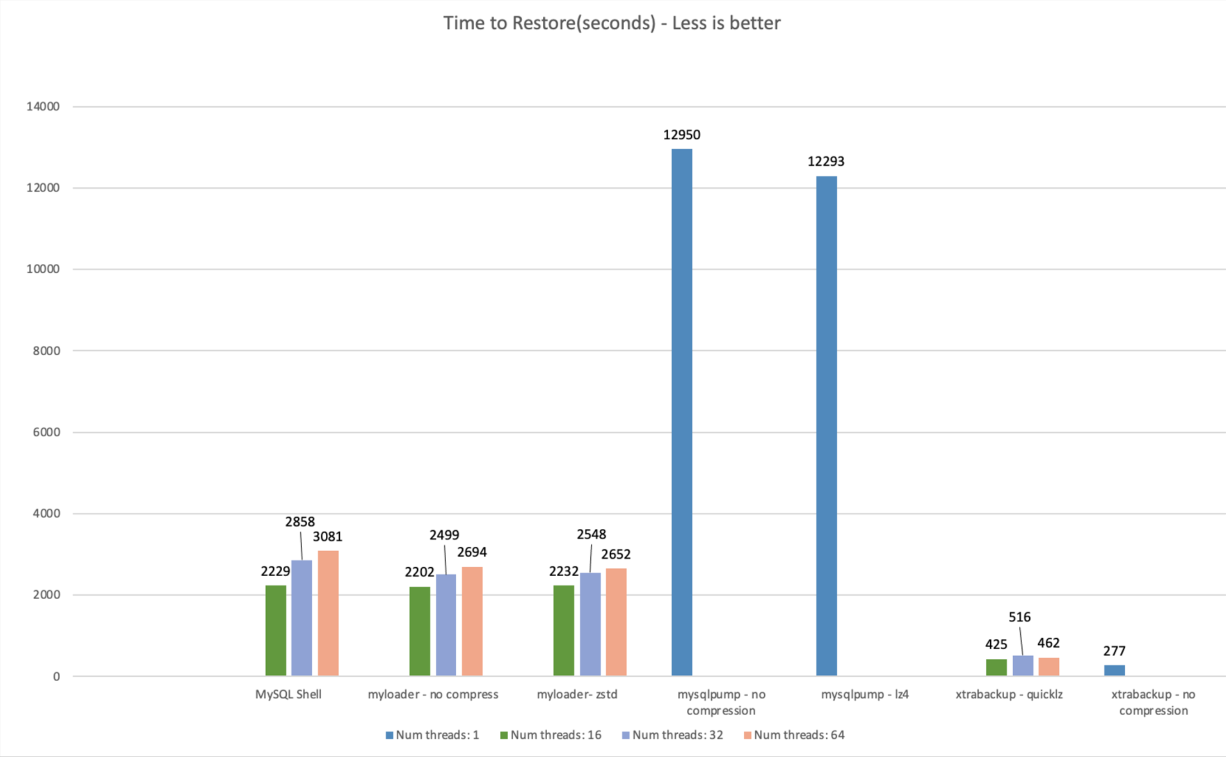
Results



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Conclusion

The final side of the history…

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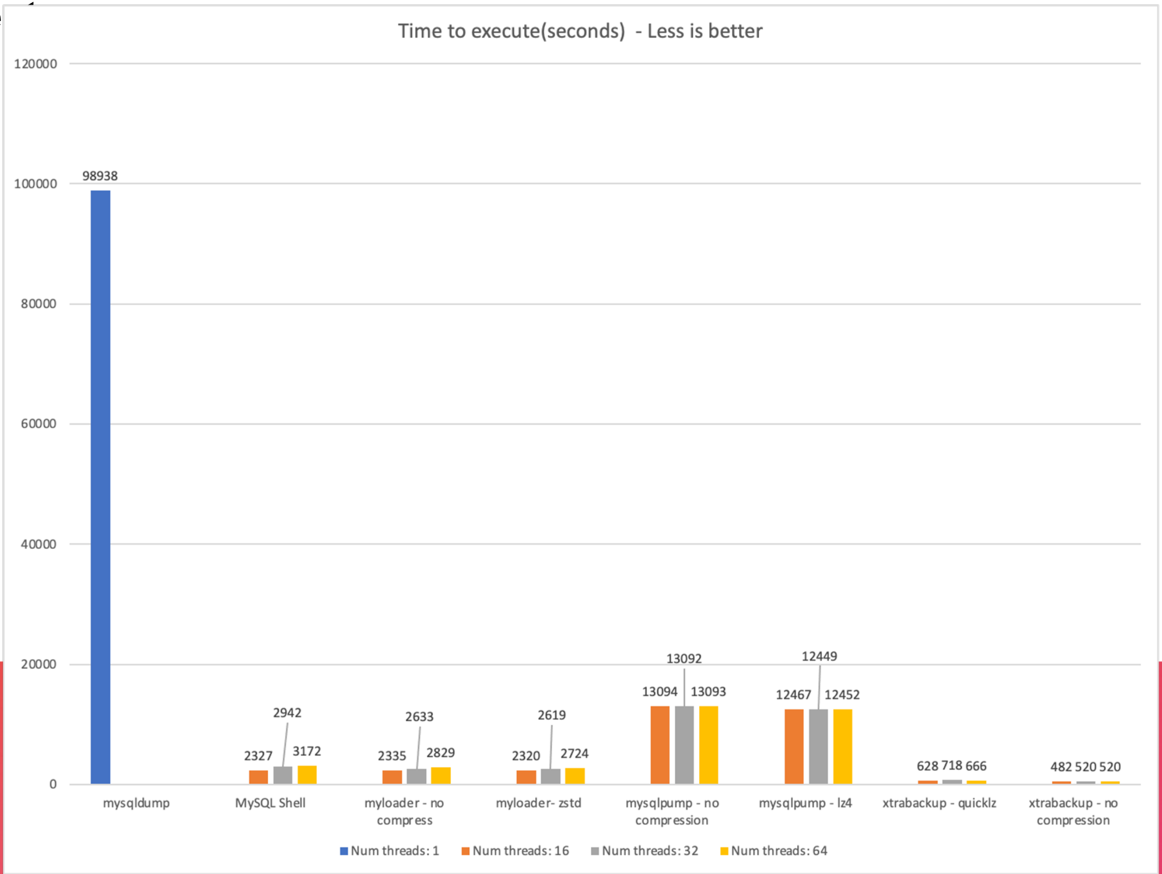
Backup + Re



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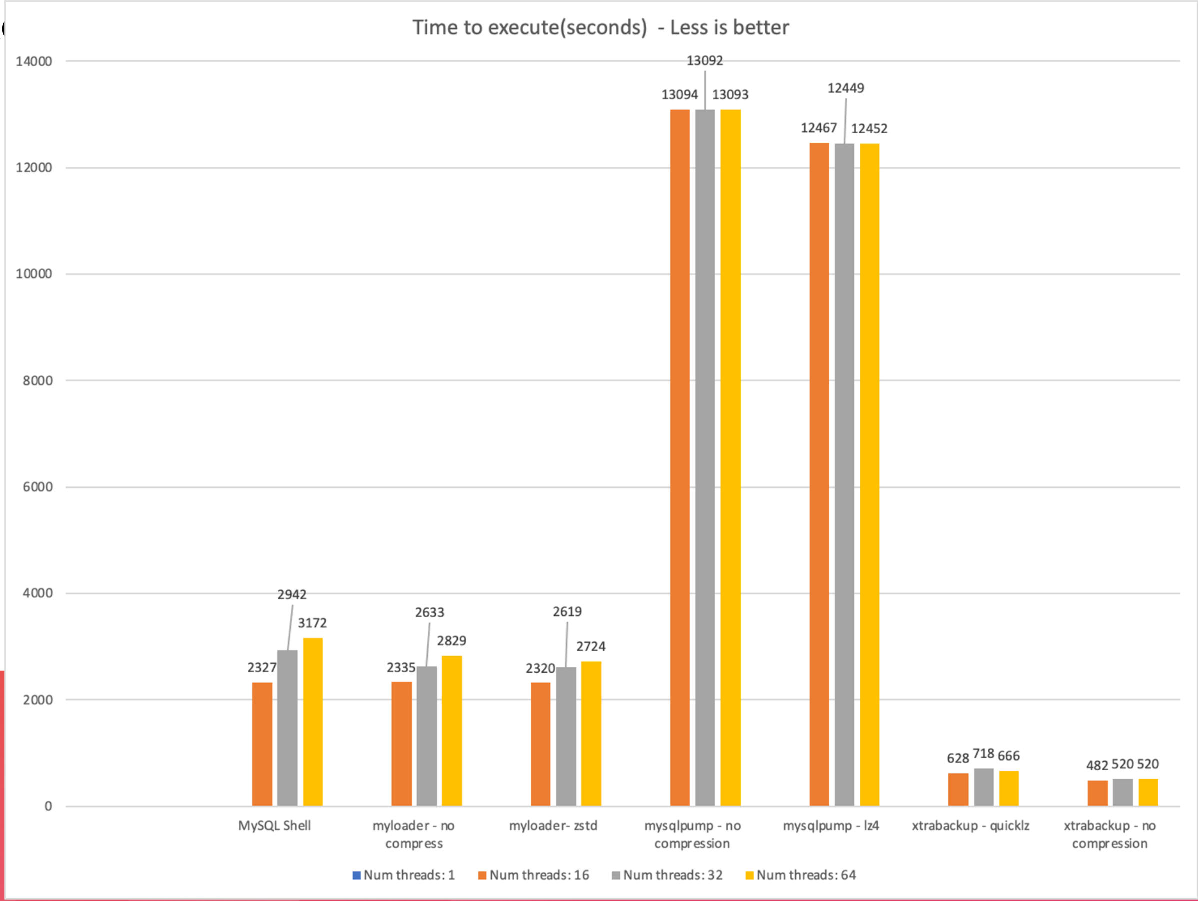
Backup + R



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Analysis

- Xtrabackup seems to offer the best balance between backup/restore time with backup size (when using compression).
- mydumper/myloader and MySQL shell are the best backup/restore options for logical backup.
- mysqlpump offers great backup capacity, but the lack of parallelism to restore the data is a great disadvantage for the tool.

Analysis

- Overall, using compression does not impact significantly the performance of backup/restore. The big advantage is disk saving.
- The parallelism offers a big boost in performance. However, the benefits of increasing the number of threads are limited by the I/O capacity.

Analysis

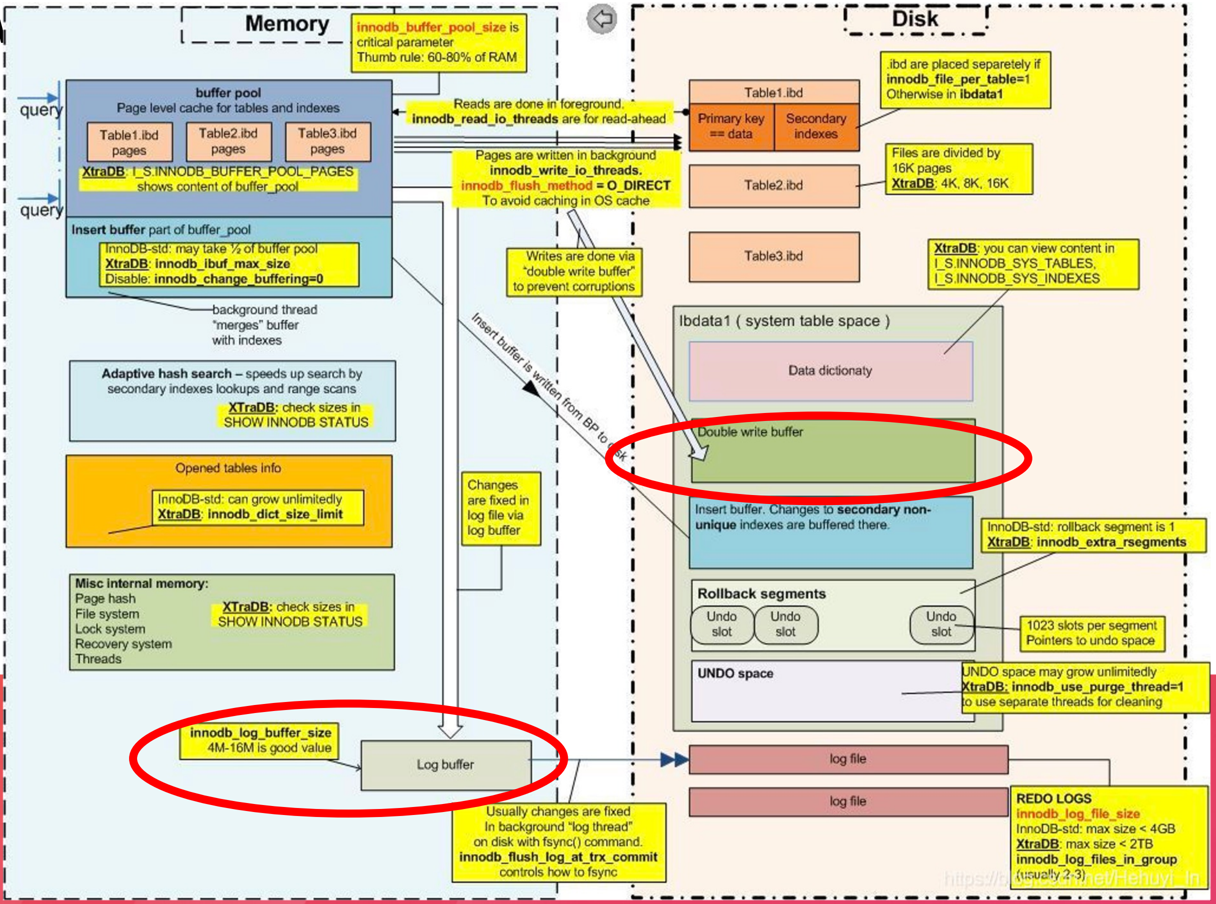
- It is possible to squeeze more juice from logical restores. Disabling writes in the binary logs, flexing ACID properties so it makes writes asynchronous, disable double write buffer are options that can be taken if you can trade reliability for performance.
- Blog post:
    - How to improve InnoDB performance by 55% for write-bound loads

Analysis

Example of settings:

```
[mysqld]
innodb_write_io_threads=8  innodb_buffer_pool_size=20G
innodb_log_file_size = X Gb #Small log files, more page flush
innodb_flush_method=O_DIRECT
innodb_flush_log_at_trx_commit=0
skip-innodb-doublewrite
```

Analysis - M



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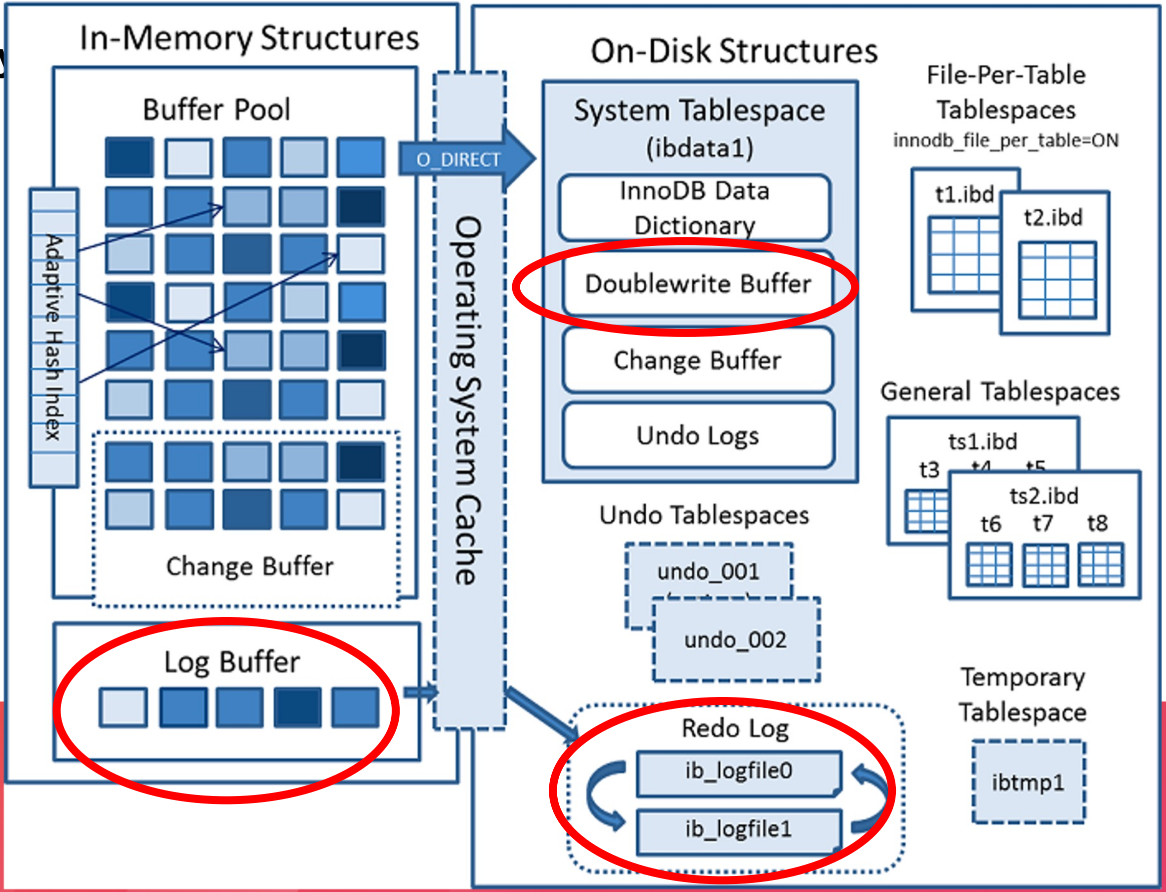
Analysis - My


In-Memory Structures
Buffer Pool
Adaptive Hash Index
Change Buffer
Log Buffer
O_DIRECT
Operating System Cache
On-Disk Structures
System Tablespace
(ibdata1)
InnoDB Data
Dictionary
Doublewrite Buffer
Change Buffer
Undo Logs
File-Per-Table
Tablespaces
innodb_file_per_table=ON
t1.ibd
t2.ibd
General Tablespaces
ts1.ibd
t3
ts2.ibd
t6
t7
t8
Undo Tablespaces
undo_001
undo_002
Redo Log
ib_logfile0
ib_logfile1
Temporary
Tablespace
ibtmp1

Disabling redo log

- As of MySQL 8.0.21, we can disable redo logging using the `ALTER INSTANCE DISABLE INNODB REDO_LOG` statement.
- This functionality is intended for loading data into a new MySQL instance. Disabling redo logging speeds up data loading by avoiding redo log writes and doublewrite buffering.

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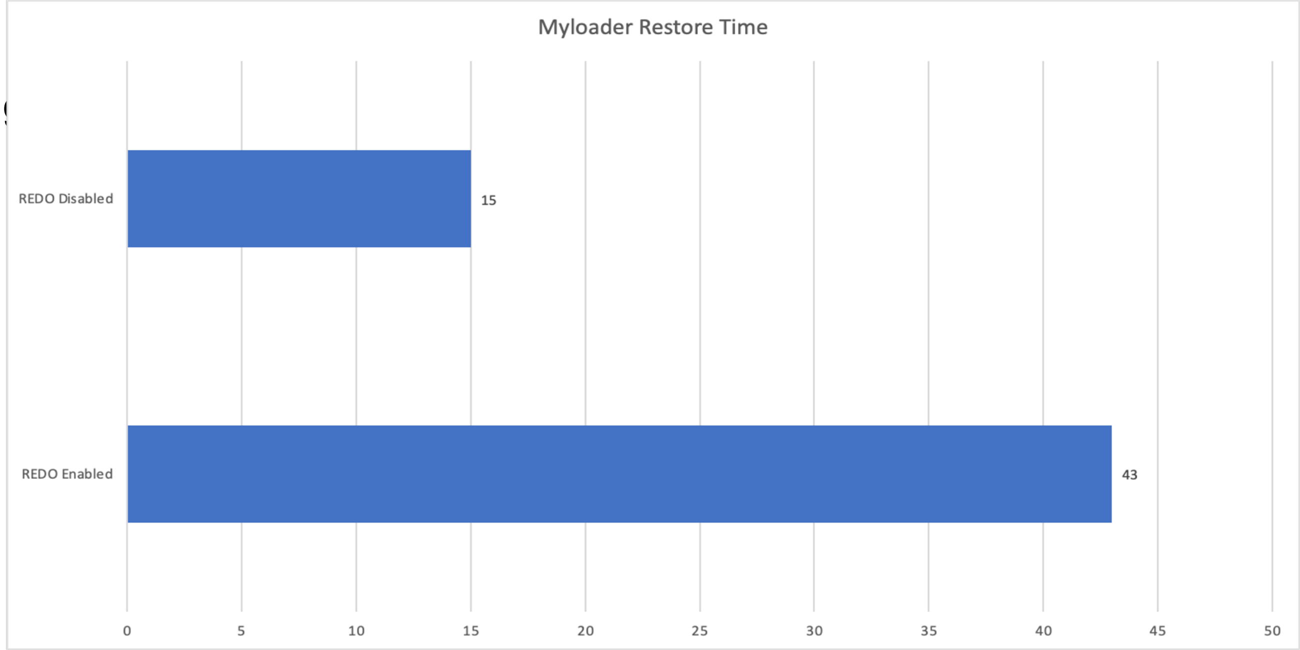
Disablin



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Choice of the presenter

Who is your favorite child?

So Vinicius, which backup and restore method would you pick? And why?

So Vinicius, which backup and restore method would you pick? And why?

- If it is a one-time backup, I would use MySQL Shell (in case of MySQL 8) or mydumper/myloader (MySQL 5.7 and 8). I'm familiar with the commands and they execute relatively fast, which is good to rebuild replica servers.
- If it is a backup/restore routine, I would go with xtrabackup. Mainly because of the PITR feature.

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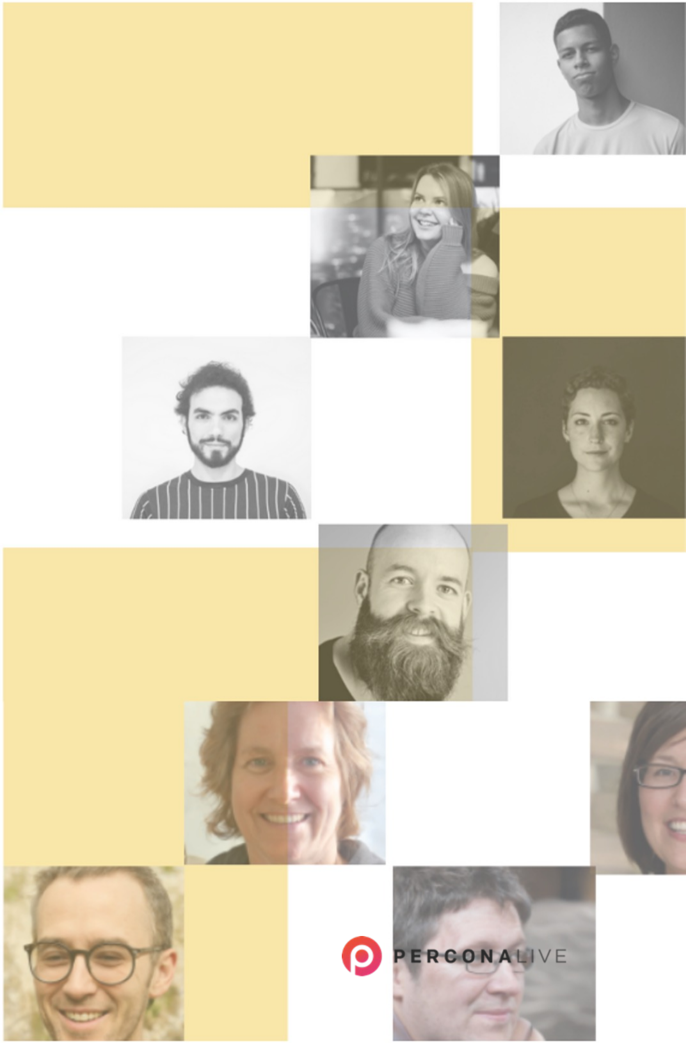

谢谢

Thank you

Grazie

Obrigado

Gracias

percona.com/contact | info@percona.com

vinicius.grippa@percona.com

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